

Synchronous Handshake Protocols and Mutual Exclusion in Shared Multi-Robot Workcells

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EXTENDED ABSTRACT

Physical part transfer between mobile robots and stationary articulated manipulators introduces acute collision hazards within shared kinematic workspaces. This paper proposes a formal synchronous handshake protocol based on ROS 2 micro-services and distributed binary semaphores to enforce strict mutual exclusion. Tested across 5,000 continuous pick-and-place transfer cycles in a physical UR5-AMR workcell, the protocol achieved zero collision events, sub-12 ms handshake response latency, and graceful error recovery during hardware disconnections.

Keywords: Mutual Exclusion, ROS 2 Services, Shared Workspace Safety, Physical Handoff, UR5, Collision Avoidance

1. Introduction & The Workspace Intersection Problem

When a mobile robot docks beside a 6-DOF industrial arm (such as a UR5), the operational reach envelopes overlap. If both robots attempt to actuate their end-effectors simultaneously due to asynchronous topic race conditions, destructive mechanical collisions occur.

The paper establishes that asynchronous topic publishing is mathematically insufficient for physical safety and demonstrates why atomic ROS 2 Services are required.

2. Handshake Protocol & State Machine

The mutual exclusion service `/acquire_transfer_lock` implements a distributed binary mutex:

```
Request: { robot_id: string, zone_id: int32, request_lock: bool }
Response: { lock_granted: bool, message: string, timestamp: int64 }
```

State transitions enforce that UR5 motion trajectories remain locked in a standby safe configuration until `lock_granted == True` is received.

3. Experimental Performance Validation

Evaluated across 5,000 real-world transfer cycles with a Universal Robots UR5 and an autonomous mobile base.

Protocol Type	Handshake Latency	Packet Loss Resilience	Collision Rate (5,000 cycles)
Unsynchronized Topics	4.2 ms	Fails on packet loss	2.4% (120 collisions)
ROS 2 Service Mutex	11.4 ms	100% Deterministic	0.0% (Zero collisions)
Centralized DB Polling	85.0 ms	High overhead	0.0% (Zero collisions)

4. Safety Integrity Analysis

- The 11.4 ms latency of ROS 2 Services provides negligible overhead while guaranteeing absolute safety.
- Automatic watchdog timers release locks if a mobile robot disconnects, preventing cell freezing.

DIRECT RELEVANCE & SYNTHESIS FOR TEAM 3 PROJECT

Directly justifies and designs Team 3's `AcquireHandoffLock.srv` and `handoff_service_server.py`, ensuring that the TurtleBot3 and UR5 interact safely without physical collisions.